

BÀI TẬP MÔN HỌC MÁY VÀ KHAI PHÁ DỮ LIỆU

Xây dựng mạng nơ-ron nhân tạo đa lớp MLP (không dùng các thư viện có sẵn) để phân lớp bộ dữ liệu [Student Performance \(file ở đây\)](#)

Lập bảng so sánh các giá trị: Thời gian, số bước lặp, sai số khi thay đổi các tham số mạng , epochs, điều kiện dừng...

Upload dạng file pdf (báo cáo) và code nguồn (file .py)

Thông tin về dataset:

1. Dữ liệu được thu thập từ sinh viên năm 2019. Mục đích là dự đoán kết quả cuối kỳ của sinh viên bằng các kỹ thuật ML.
2. Cột từ 1-10 là câu hỏi cá nhân, 11-16 câu hỏi bao gồm câu hỏi về gia đình và các câu hỏi còn lại bao gồm thói quen giáo dục.

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
Student Age	Feature	Categorical	Age	1: 18-21, 2: 22-25, 3: above 26		no
Sex	Feature	Binary	Sex	1: female, 2: male		no
Graduated high-school type	Feature	Categorical	Education Level	1: private, 2: state, 3: other		no
Scholarship type	Feature	Categorical		1: None, 2: 25%, 3: 50%, 4: 75%, 5: Full		no
Additional work	Feature	Binary		1: Yes, 2: No		no
Regular artistic or sports activity	Feature	Binary		1: Yes, 2: No		no
Do you have a partner	Feature	Binary	Marital Status	1: Yes, 2: No		no
Total salary if available	Feature	Categorical	Income	1: USD 135-200, 2: USD 201-270, 3:		no

				USD 271-340, 4: USD 341-410, 5: above 410		
Transportation to the university	Feature	Categorical		1: Bus, 2: Private car/taxi, 3: bicycle, 4: Other		no
Accommodation type in Cyprus	Feature	Categorical		1: rental, 2: dormitory, 3: with family, 4: Other		no
Mother's education	Feature	Categorical	Education Level	1: primary school, 2: secondary school, 3: high school, 4: university, 5: MSc., 6: Ph.D.		no
Father's education	Feature	Categorical	Education Level	1: primary school, 2: secondary school, 3: high school, 4: university, 5: MSc., 6: Ph.D.		no
Number of sisters/brothers (if available)	Feature	Integer		1: 1, 2: 2, 3: 3, 4: 4, 5: 5 or above		no
Parental status	Feature	Categorical	Marital Status	1: married, 2: divorced, 3:		no

				died - one of them or both		
Mother's occupation	Feature	Categorical	Occupation	1: retired, 2: housewife, 3: government officer, 4: private sector employee, 5: self-employment, 6: other		no
Father's occupation	Feature	Categorical	Occupation	1: retired, 2: housewife, 3: government officer, 4: private sector employee, 5: self-employment, 6: other		no
Weekly study hours	Feature	Categorical		1: None, 2: <5 hours, 3: 6-10 hours, 4: 11-20 hours, 5: more than 20 hours		no
Reading frequency (non-scientific books/journals)	Feature	Categorical		1: None, 2: Sometimes, 3: Often		no
Reading frequency (scientific books/journals)	Feature	Categorical		1: None, 2: Sometimes, 3: Often		no
Attendance to the seminars/conferences	Feature	Binary		1: Yes, 2: No		no

related to the department						
Impact of your projects/activities on your success	Feature	Categorical		1: positive, 2: negative, 3: neutral		no
Attendance to classes	Feature	Categorical		1: always, 2: sometimes, 3: never		no
Preparation to midterm exams 1	Feature	Categorical		1: alone, 2: with friends, 3: not applicable		no

Class Labels

Student ID

- 1- Student Age (1: 18-21, 2: 22-25, 3: above 26)
- 2- Sex (1: female, 2: male)
- 3- Graduated high-school type: (1: private, 2: state, 3: other)
- 4- Scholarship type: (1: None, 2: 25%, 3: 50%, 4: 75%, 5: Full)
- 5- Additional work: (1: Yes, 2: No)
- 6- Regular artistic or sports activity: (1: Yes, 2: No)
- 7- Do you have a partner: (1: Yes, 2: No)
- 8- Total salary if available (1: USD 135-200, 2: USD 201-270, 3: USD 271-340, 4: USD 341-410, 5: above 410)
- 9- Transportation to the university: (1: Bus, 2: Private car/taxi, 3: bicycle, 4: Other)
- 10- Accommodation type in Cyprus: (1: rental, 2: dormitory, 3: with family, 4: Other)
- 11- Mothers education: (1: primary school, 2: secondary school, 3: high school, 4: university, 5: MSc., 6: Ph.D.)
- 12- Fathers education: (1: primary school, 2: secondary school, 3: high school, 4: university, 5: MSc., 6: Ph.D.)
- 13- Number of sisters/brothers (if available): (1: 1, 2: 2, 3: 3, 4: 4, 5: 5 or above)
- 14- Parental status: (1: married, 2: divorced, 3: died - one of them or both)
- 15- Mothers occupation: (1: retired, 2: housewife, 3: government officer, 4: private sector employee, 5: self-employment, 6: other)
- 16- Fathers occupation: (1: retired, 2: government officer, 3: private sector employee, 4: self-employment, 5: other)
- 17- Weekly study hours: (1: None, 2: <5 hours, 3: 6-10 hours, 4: 11-20 hours, 5: more than 20 hours)

20 hours)

18- Reading frequency (non-scientific books/journals): (1: None, 2: Sometimes, 3: Often)

19- Reading frequency (scientific books/journals): (1: None, 2: Sometimes, 3: Often)

20- Attendance to the seminars/conferences related to the department: (1: Yes, 2: No)

21- Impact of your projects/activities on your success: (1: positive, 2: negative, 3: neutral)

22- Attendance to classes (1: always, 2: sometimes, 3: never)

23- Preparation to midterm exams 1: (1: alone, 2: with friends, 3: not applicable)

24- Preparation to midterm exams 2: (1: closest date to the exam, 2: regularly during the semester, 3: never)

25- Taking notes in classes: (1: never, 2: sometimes, 3: always)

26- Listening in classes: (1: never, 2: sometimes, 3: always)

27- Discussion improves my interest and success in the course: (1: never, 2: sometimes, 3: always)

28- Flip-classroom: (1: not useful, 2: useful, 3: not applicable)

29- Cumulative grade point average in the last semester (/4.00): (1: <2.00, 2: 2.00-2.49, 3: 2.50-2.99, 4: 3.00-3.49, 5: above 3.49)

30- Expected Cumulative grade point average in the graduation (/4.00): (1: <2.00, 2: 2.00-2.49, 3: 2.50-2.99, 4: 3.00-3.49, 5: above 3.49)

31- Course ID

32- OUTPUT Grade (0: Fail, 1: DD, 2: DC, 3: CC, 4: CB, 5: BB, 6: BA, 7: AA)